

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from google.colab import files
uploaded = files.upload()
```

No file chosen

```
df = pd.read_csv("AEP_hourly.csv")
```

```
df.head()
```

	Datetime	AEP_MW
0	2004-12-31 01:00:00	13478.0
1	2004-12-31 02:00:00	12865.0
2	2004-12-31 03:00:00	12577.0
3	2004-12-31 04:00:00	12517.0
4	2004-12-31 05:00:00	12670.0

```
df.columns
```

```
Index(['Datetime', 'AEP_MW'], dtype='object')
```

```
df.isnull().sum()
```

	0
Datetime	0
AEP_MW	0

dtype: int64

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 121273 entries, 0 to 121272
Data columns (total 2 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Datetime    121273 non-null object
1   AEP_MW      121273 non-null float64
dtypes: float64(1), object(1)
memory usage: 1.9+ MB
```

```
df['Datetime'] = pd.to_datetime(df['Datetime'])
```

```
df['Hour'] = df['Datetime'].dt.hour
df['Day'] = df['Datetime'].dt.day
df['Month'] = df['Datetime'].dt.month
df['Year'] = df['Datetime'].dt.year
df['Day_Name'] = df['Datetime'].dt.day_name()
```

```
df.rename(columns={'AEP_MW': 'Energy_Consumption'}, inplace=True)
```

```
df.head()
```

	Datetime	Energy_Consumption	Hour	Day	Month	Year	Day_Name
0	2004-12-31 01:00:00	13478.0	1	31	12	2004	Friday
1	2004-12-31 02:00:00	12865.0	2	31	12	2004	Friday
2	2004-12-31 03:00:00	12577.0	3	31	12	2004	Friday
3	2004-12-31 04:00:00	12517.0	4	31	12	2004	Friday
4	2004-12-31 05:00:00	12670.0	5	31	12	2004	Friday

```
total_energy = df['Energy_Consumption'].sum()
print("Total Energy Consumption:", total_energy)
```

Total Energy Consumption: 1879672527.0

```
avg_energy = df['Energy_Consumption'].mean()
print("Average Energy Consumption:", avg_energy)
```

Average Energy Consumption: 15499.513716985644

```
max_energy = df['Energy_Consumption'].max()
print("Peak Energy Consumption:", max_energy)
```

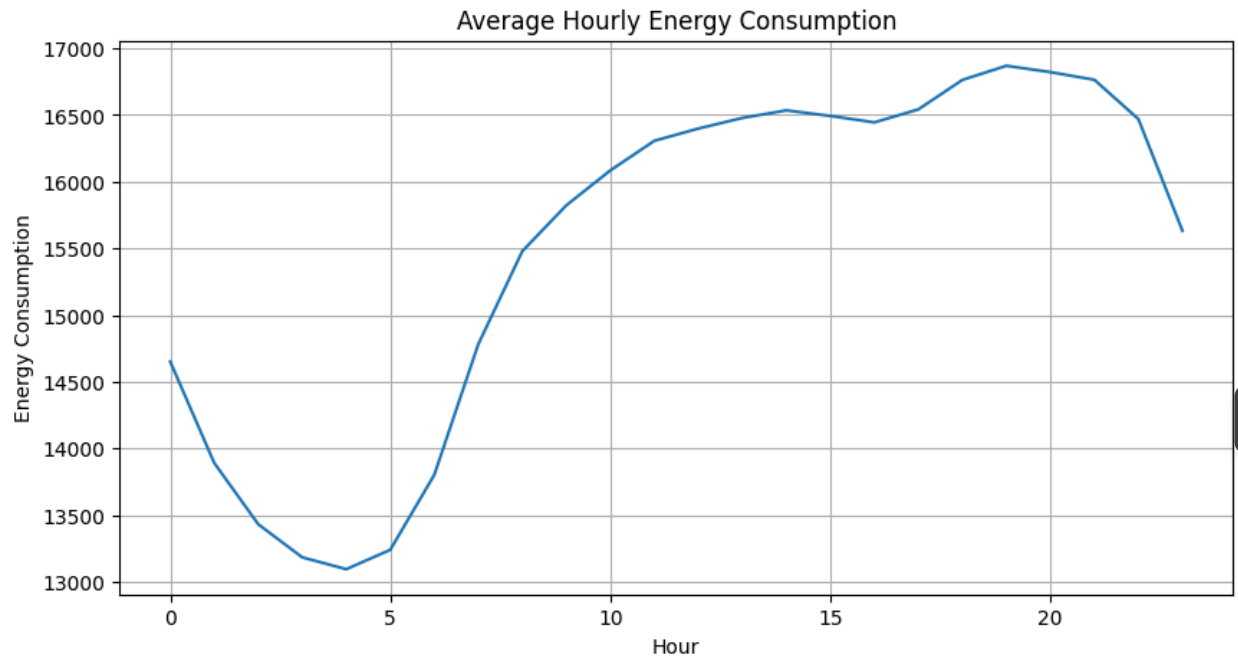
Peak Energy Consumption: 25695.0

```
hourly_usage = df.groupby('Hour')['Energy_Consumption'].mean()
hourly_usage
```

Energy_Consumption	
Hour	
0	14651.191569
1	13891.478433
2	13432.062995
3	13184.049008
4	13095.193350
5	13240.535813
6	13802.401464
7	14781.668381
8	15478.830233
9	15822.653740
10	16084.283934
11	16306.315592
12	16398.177087
13	16477.793035
14	16534.660202
15	16492.810645
16	16444.506134
17	16542.038781
18	16762.940047
19	16868.728334
20	16821.335180
21	16763.806292
22	16468.967550
23	15632.594183

dtype: float64

```
plt.figure(figsize=(10,5))
hourly_usage.plot()
plt.title("Average Hourly Energy Consumption")
plt.xlabel("Hour")
plt.ylabel("Energy Consumption")
plt.grid()
plt.show()
```

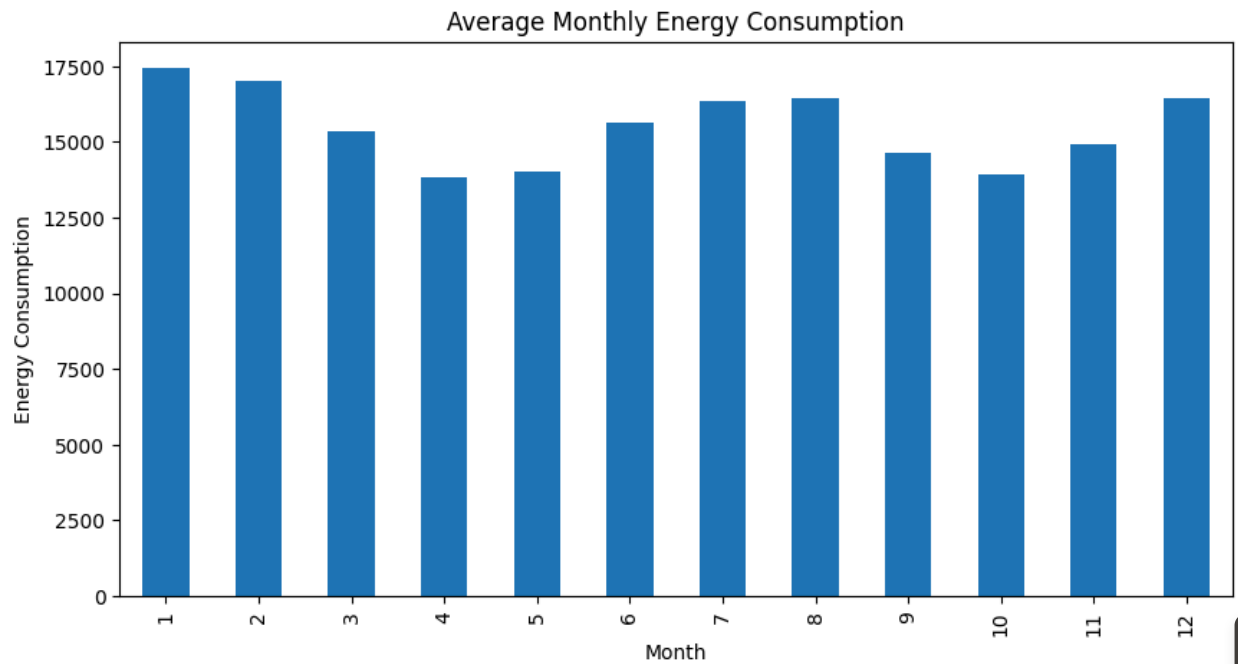


```
monthly_usage = df.groupby('Month')['Energy_Consumption'].mean()
monthly_usage
```

Energy_Consumption	
Month	
1	17431.269009
2	17022.815084
3	15376.835720
4	13823.857511
5	14006.393817
6	15630.021825
7	16349.853303
8	16424.881699
9	14657.280556
10	13938.926431
11	14930.405676
12	16445.609084

dtype: float64

```
plt.figure(figsize=(10,5))
monthly_usage.plot(kind='bar')
plt.title("Average Monthly Energy Consumption")
plt.xlabel("Month")
plt.ylabel("Energy Consumption")
plt.show()
```



```
day_usage = df.groupby('Day_Name')['Energy_Consumption'].mean()
```

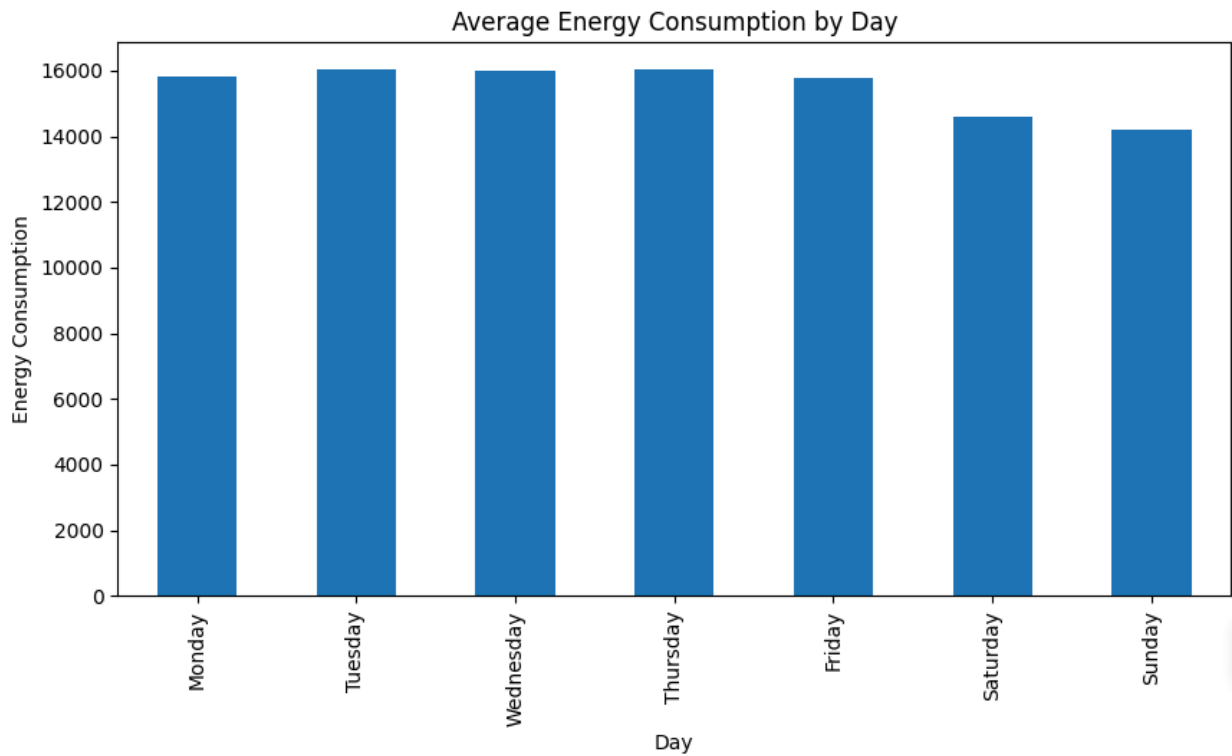
```
day_usage = day_usage.reindex([
    'Monday', 'Tuesday', 'Wednesday',
    'Thursday', 'Friday', 'Saturday', 'Sunday'
])
```

```
day_usage
```

Energy_Consumption	
Day_Name	
Monday	15810.973684
Tuesday	16057.615571
Wednesday	16013.589739
Thursday	16028.138281
Friday	15773.123911
Saturday	14610.979628
Sunday	14200.754680

```
dtype: float64
```

```
plt.figure(figsize=(10,5))
day_usage.plot(kind='bar')
plt.title("Average Energy Consumption by Day")
plt.xlabel("Day")
plt.ylabel("Energy Consumption")
plt.show()
```



```
peak_hour = hourly_usage.idxmax()

print("Peak Usage Hour:", peak_hour)
```

Peak Usage Hour: 19

```
low_hour = hourly_usage.idxmin()

print("Lowest Usage Hour:", low_hour)
```

Lowest Usage Hour: 4

```
threshold = df['Energy_Consumption'].mean() + 2*df['Energy_Consumption'].std()
```

```
anomalies = df[df['Energy_Consumption'] > threshold]

anomalies.head()
```

	Datetime	Energy_Consumption	Hour	Day	Month	Year	Day_Name
270	2004-12-20 07:00:00	21616.0	7	20	12	2004	Monday
271	2004-12-20 08:00:00	22554.0	8	20	12	2004	Monday
272	2004-12-20 09:00:00	22577.0	9	20	12	2004	Monday
273	2004-12-20 10:00:00	22321.0	10	20	12	2004	Monday
274	2004-12-20 11:00:00	22055.0	11	20	12	2004	Monday

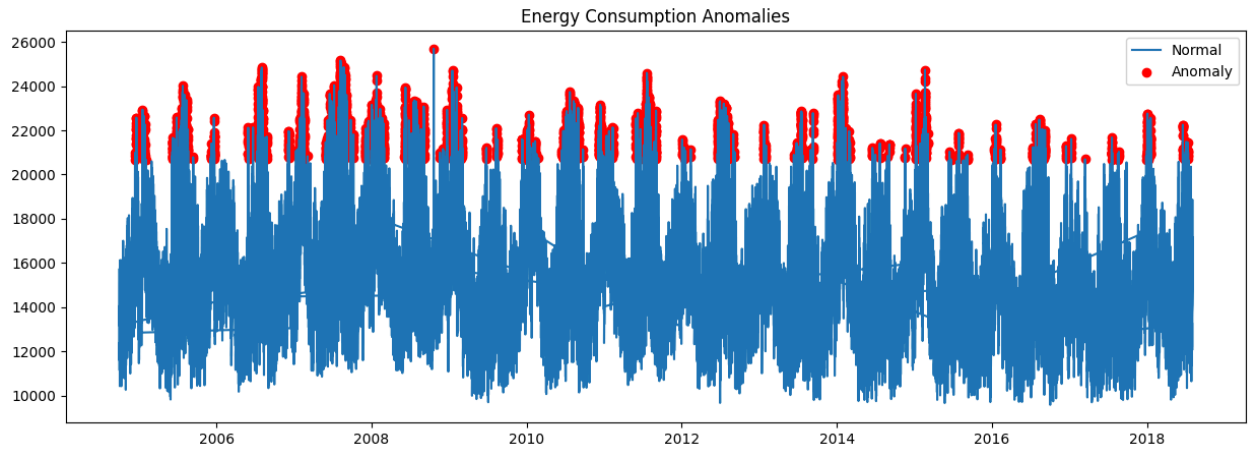
Next steps: [Generate code with anomalies](#) [New interactive sheet](#)

```
plt.figure(figsize=(15,5))

plt.plot(df['Datetime'], df['Energy_Consumption'], label='Normal')

plt.scatter(anomalies['Datetime'],
            anomalies['Energy_Consumption'],
            color='red',
            label='Anomaly')

plt.legend()
plt.title("Energy Consumption Anomalies")
plt.show()
```



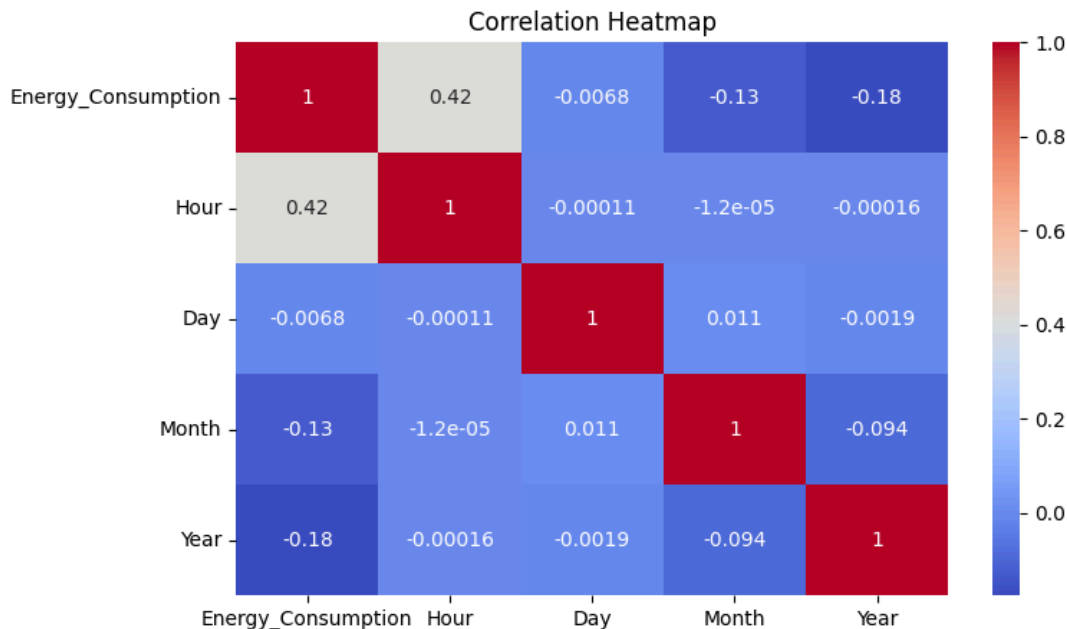
```
df.corr(numeric_only=True)
```

	Energy_Consumption	Hour	Day	Month	Year
Energy_Consumption	1.000000	0.419480	-0.006847	-0.125545	-0.175130
Hour	0.419480	1.000000	-0.000108	-0.000012	-0.000165
Day	-0.006847	-0.000108	1.000000	0.010507	-0.001892
Month	-0.125545	-0.000012	0.010507	1.000000	-0.093532
Year	-0.175130	-0.000165	-0.001892	-0.093532	1.000000

```
plt.figure(figsize=(8,5))

sns.heatmap(df.corr(numeric_only=True),
            annot=True,
            cmap='coolwarm')

plt.title("Correlation Heatmap")
plt.show()
```



```
df.to_csv("cleaned_energy_data.csv", index=False)
```

```
import sqlite3
```

```
conn = sqlite3.connect("energy.db")
```

```
df.to_sql("energy_data", conn, if_exists='replace', index=False)
```

121273

```
query = """
SELECT Month,
AVG(Energy_Consumption) as avg_consumption
FROM energy_data
GROUP BY Month
ORDER BY Month
"""
```

```
pd.read_sql(query, conn)
```

	Month	avg_consumption
0	1	17431.269009
1	2	17022.815084
2	3	15376.835720
3	4	13823.857511
4	5	14006.393817
5	6	15630.021825
6	7	16349.853303
7	8	16424.881699
8	9	14657.280556
9	10	13938.926431
10	11	14930.405676
11	12	16445.609084

```
query = """
SELECT Hour,
AVG(Energy_Consumption) as avg_consumption
FROM energy_data
GROUP BY Hour
ORDER BY avg_consumption DESC
"""
```

```
pd.read_sql(query, conn)
```

	Hour	avg_consumption
0	19	16868.728334
1	20	16821.335180
2	21	16763.806292
3	18	16762.940047
4	17	16542.038781
5	14	16534.660202

```
query = """
SELECT Day_Name,
AVG(Energy_Consumption) as avg_consumption
FROM energy_data
GROUP BY Day_Name
"""

pd.read_sql(query, conn)
```

	Day_Name	avg_consumption
0	Friday	15773.123911
1	Monday	15810.973684
2	Saturday	14610.979628
3	Sunday	14200.754680
4	Thursday	16028.138281

